SPB1000367-REV.1-MF1A-BR00

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Revision History

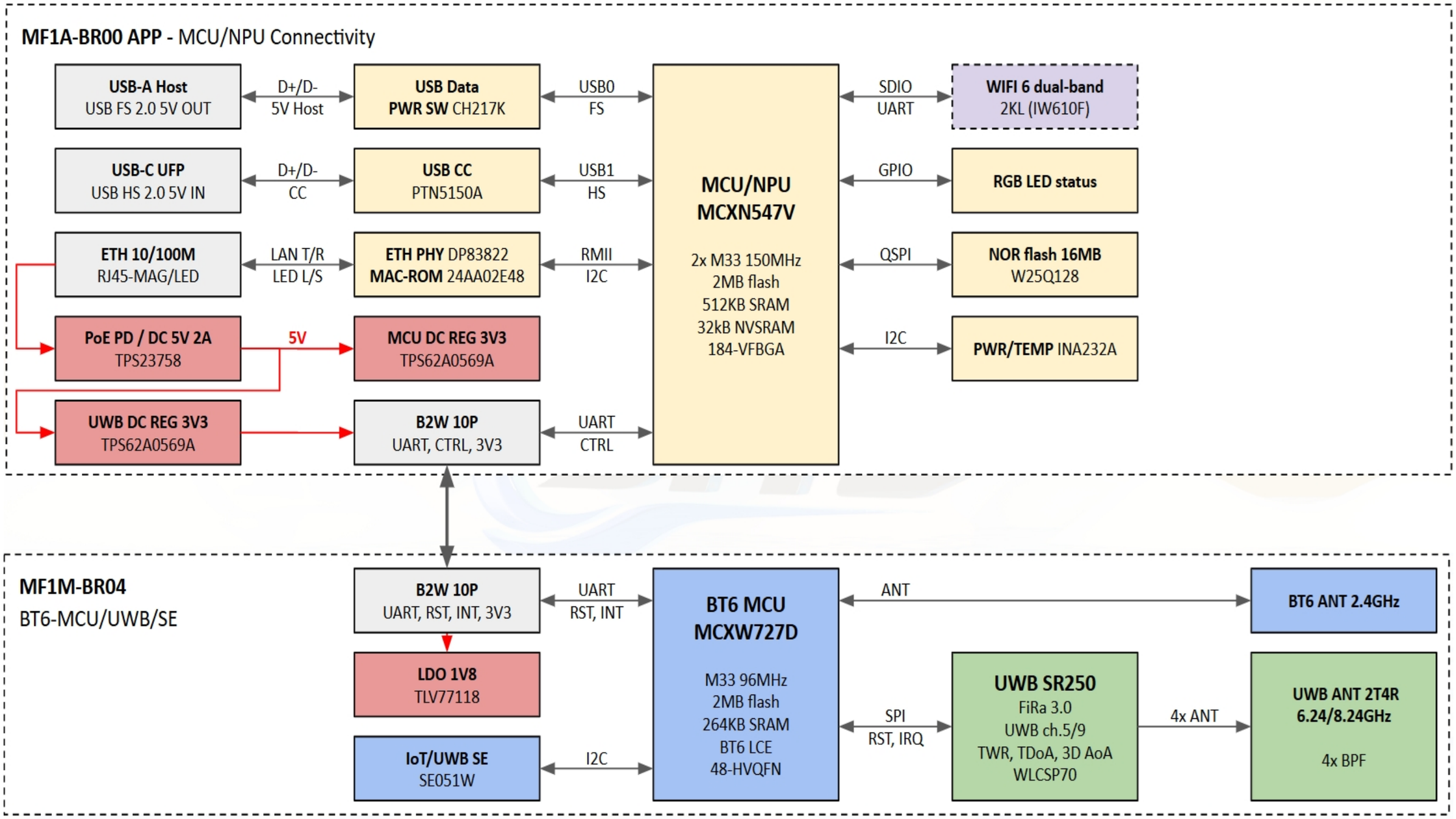
Rev	Date	Description
REV.1	04-NOV-2026	Initial Release

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TITLE MF1A-BR00		DWG NO: SPB1000367	
DRAWN BY: THAI LAM	COVER PAGE	SIZE A3	REV 1
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MF1A-BR00 block diagram

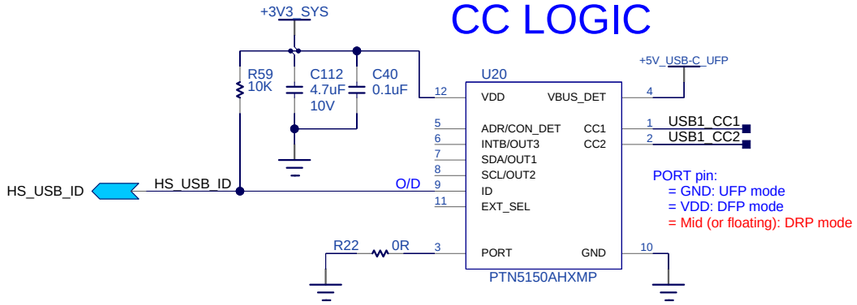


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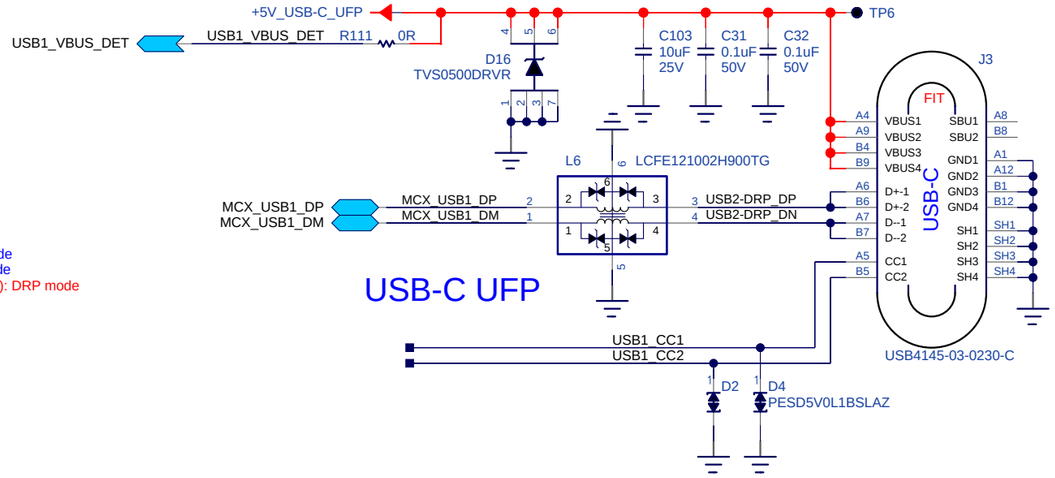
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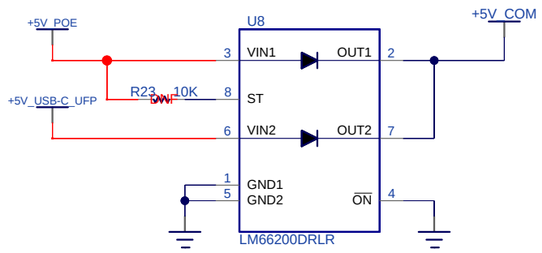
CC LOGIC



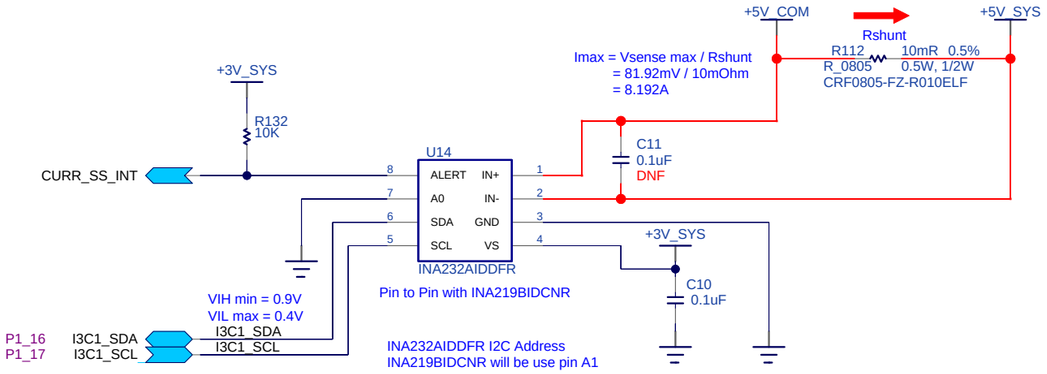
USB-C UFP



ORING IDEAL DIODE



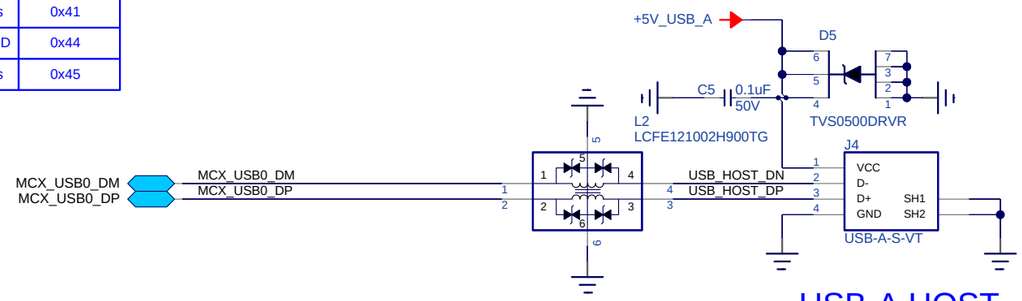
CURRENT/VOLTAGE SENSOR



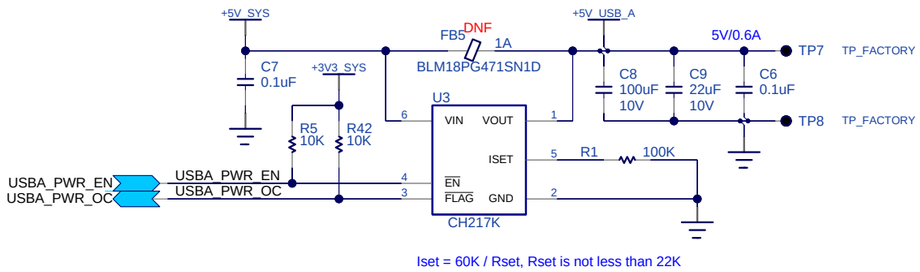
INA232AIDDFR I2C Address
INA219BIDCNR will be use pin A1

A1	A0	I2C Address
GND	GND	0x40
GND	Vs	0x41
Vs	GND	0x44
Vs	Vs	0x45

USB-A HOST



VBUS PWR CTRL



Iset = 60K / Rset, Rset is not less than 22K

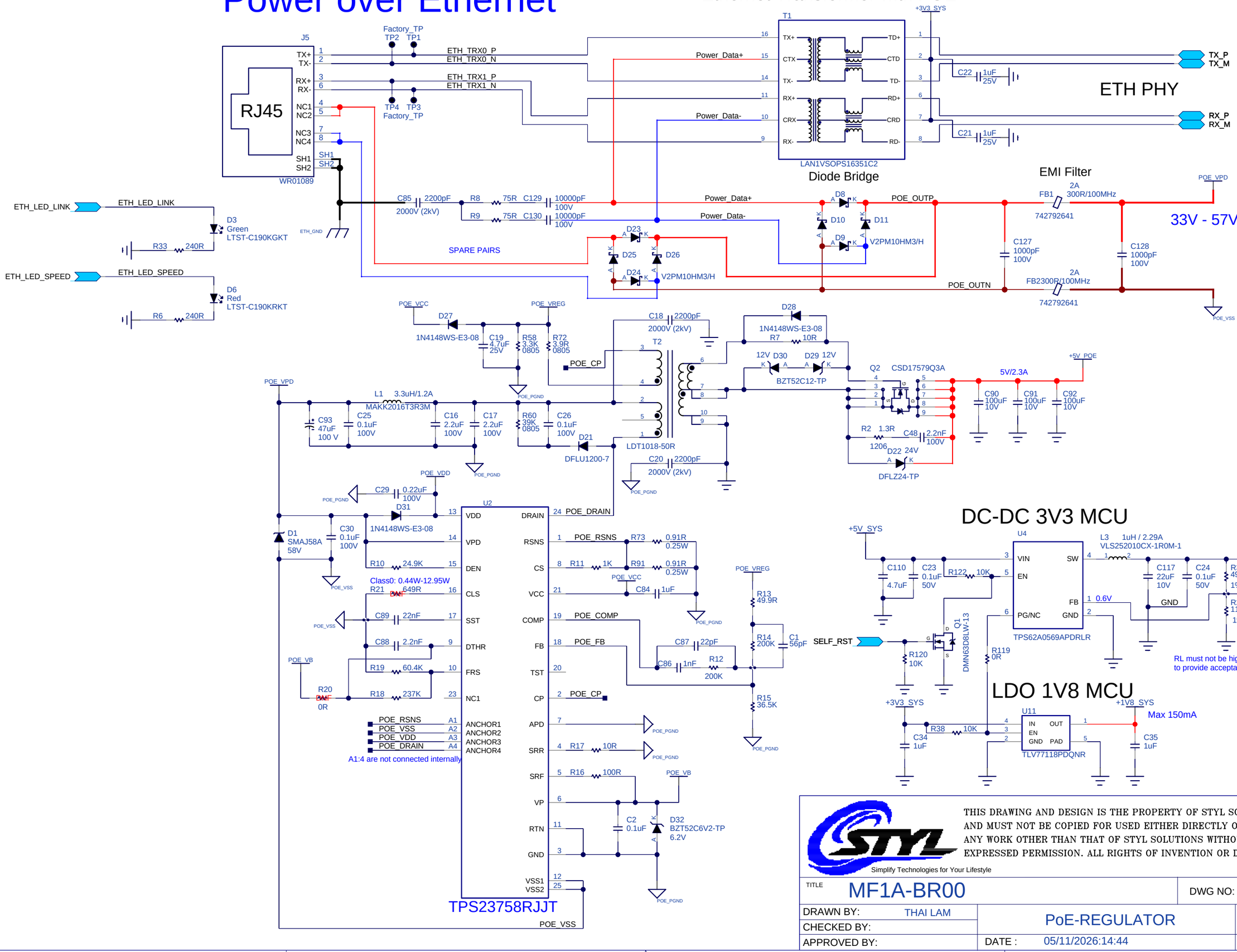
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Power over Ethernet

Ethernet Transformer with POE

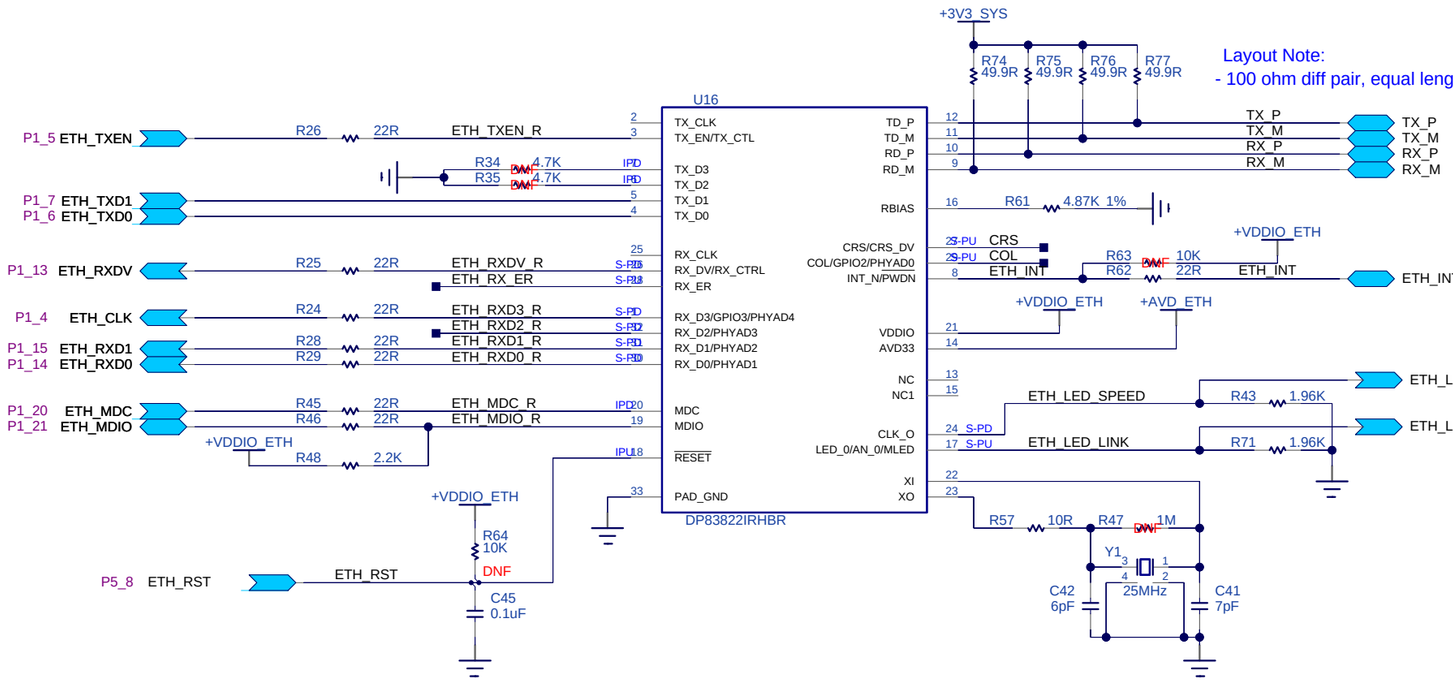


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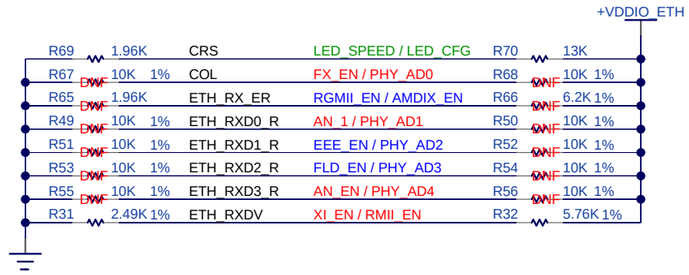
TITLE: MF1A-BR00		DWG NO: SPB1000367	
DRAWN BY: THAI LAM	PoE-REGULATOR	SIZE A3	REV 1
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ETHERNET 100M
RMII Leader signaling



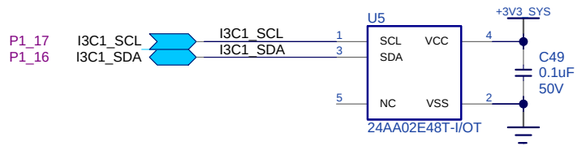
PHY Strap pins config

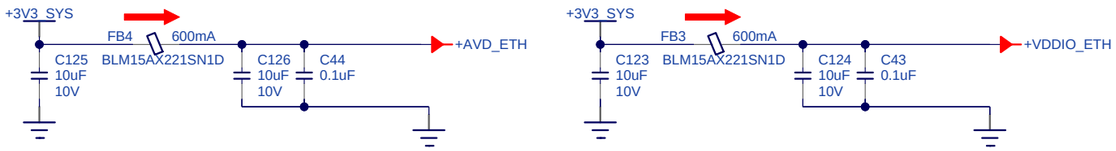
Configuration bootstraps refer Table 8-10. 4-Level Strap Pins in DP83822 Datasheet



COL, LED_0, CRS and RX_ER have internal Pullup resistors
All other pins with bootstraps have internal Pulldown resistors

EEPROM

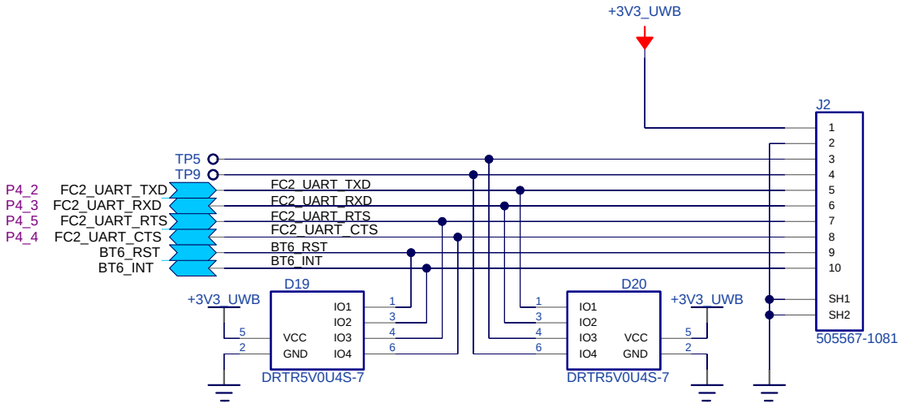




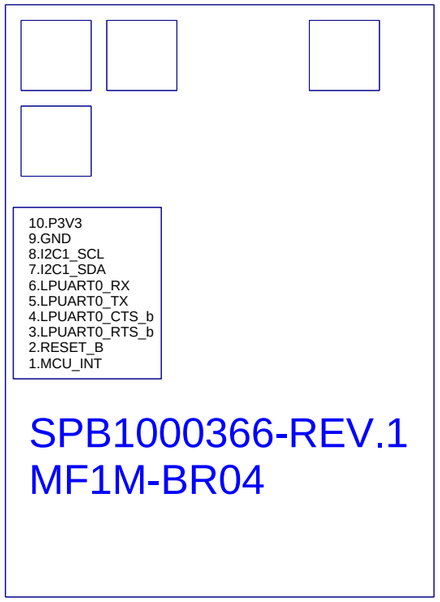
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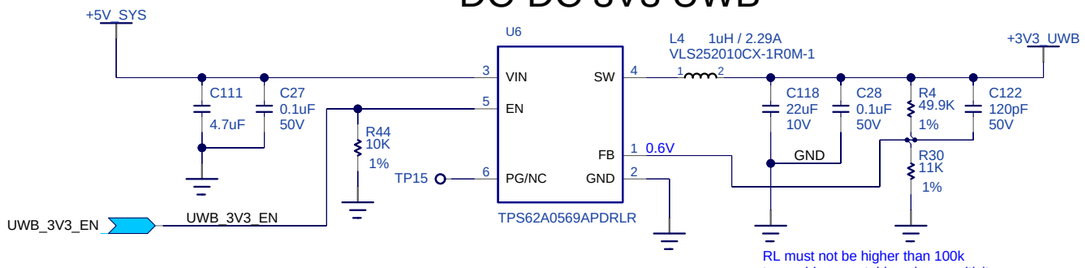


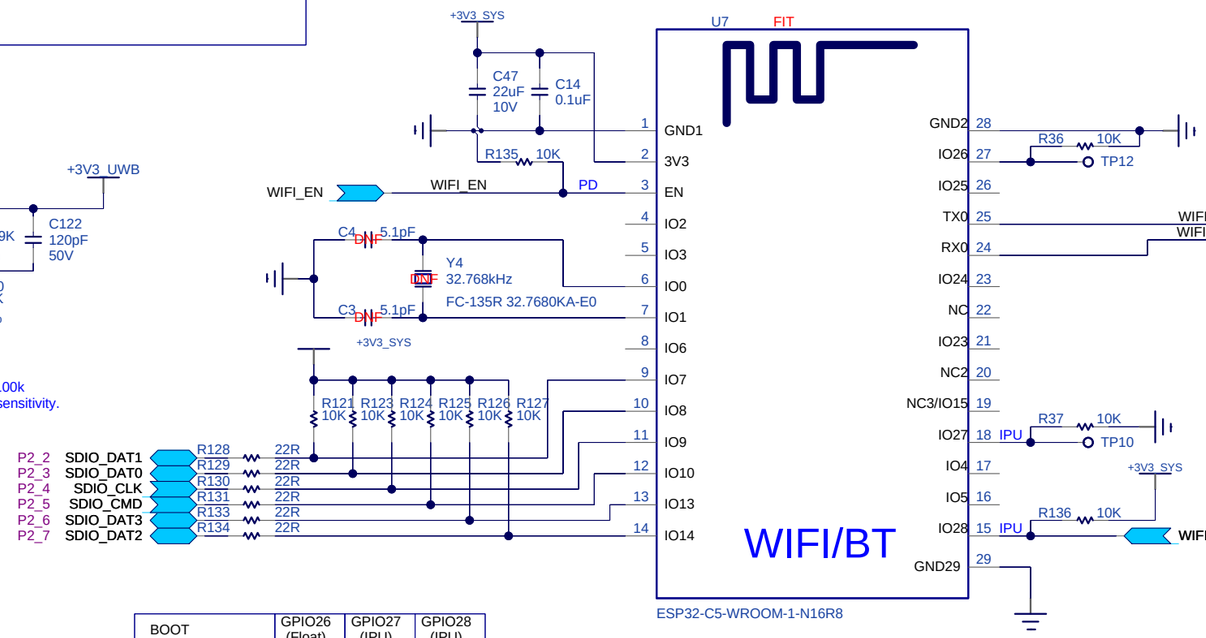
UWB B2W



SPB1000366-REV.1
MF1M-BR04

DC-DC 3V3 UWB





WIFI/BT

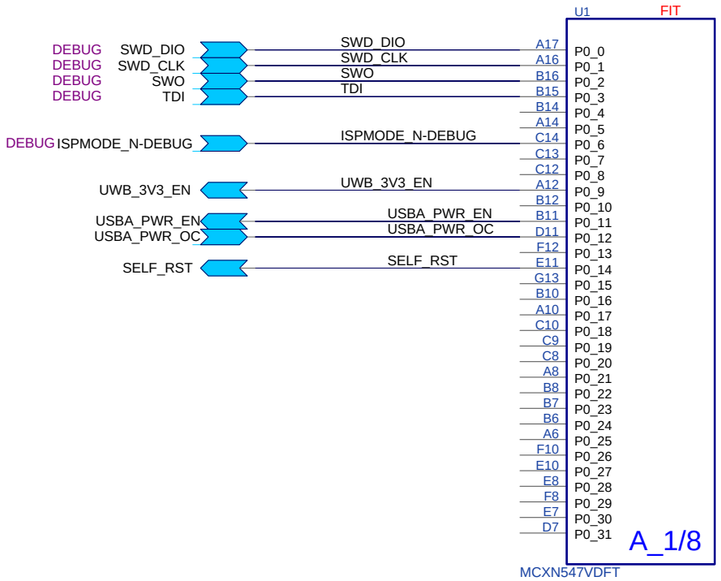
BOOT	GPIO26 (Float)	GPIO27 (IPU)	GPIO28 (IPU)
SPI BOOT(default)	ANY	ANY	1
UART/SDIO	0	0	0

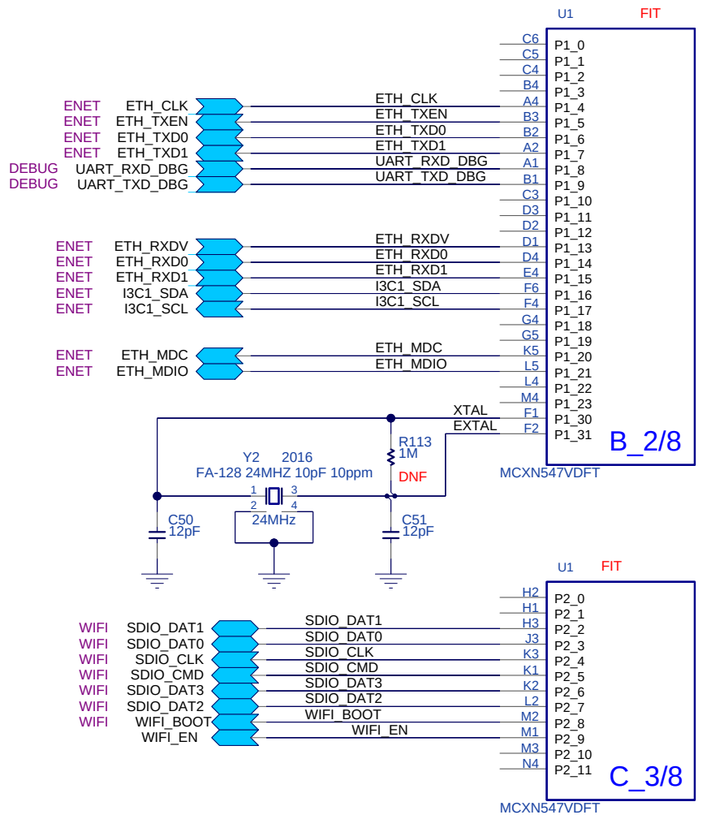
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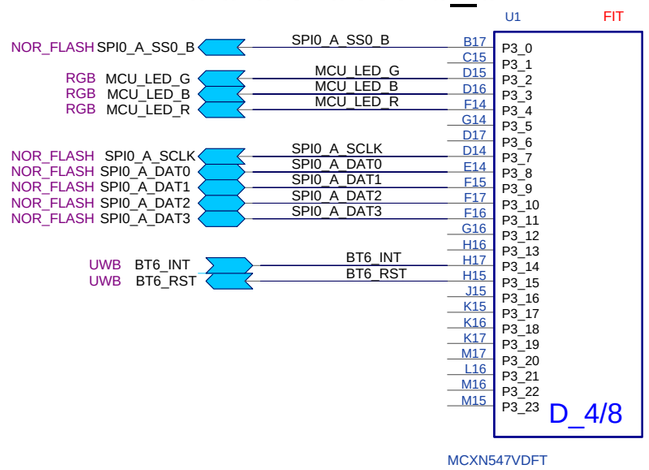
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DRAWN BY: THAI LAM	UWB_WIFI	SIZE A3	REV 1
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APPROVED BY:	DATE : 05/11/2026:15:29	SHEET 6 of 9	

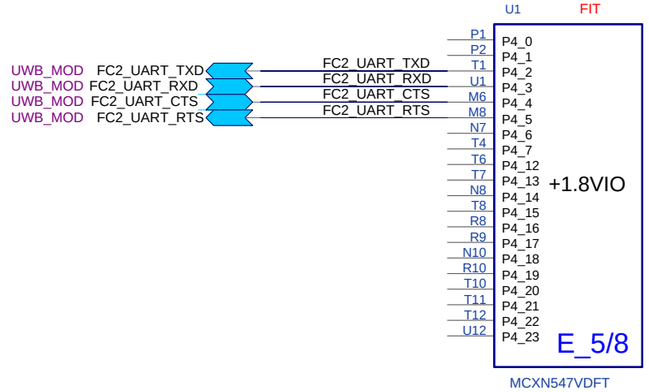
SOC PORT 0_2

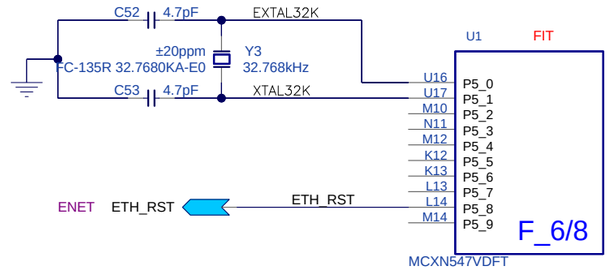




SOC PORT 3_5





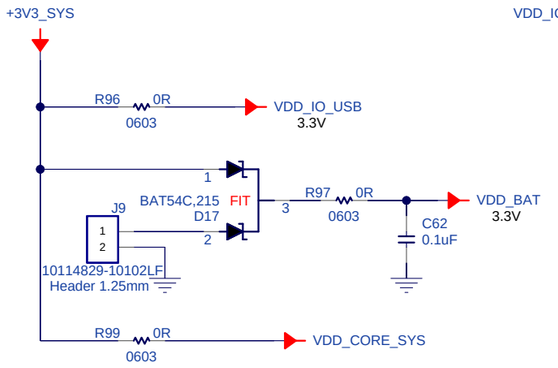


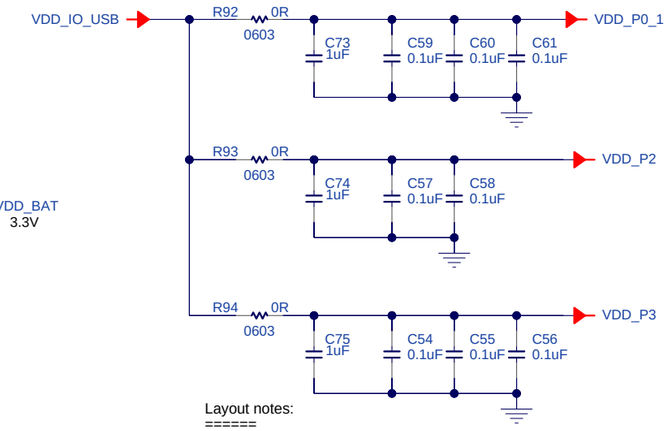
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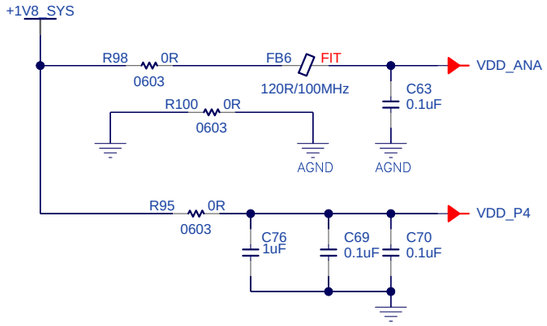
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DRAWN BY: THAI LAM	MCU_N54_PORT0-5	SIZE A3	REV 1
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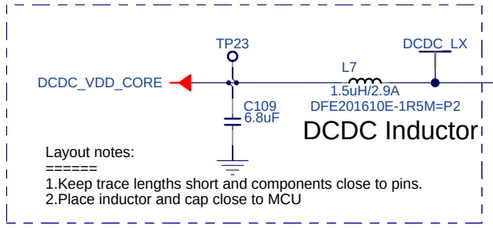
SOC POWER



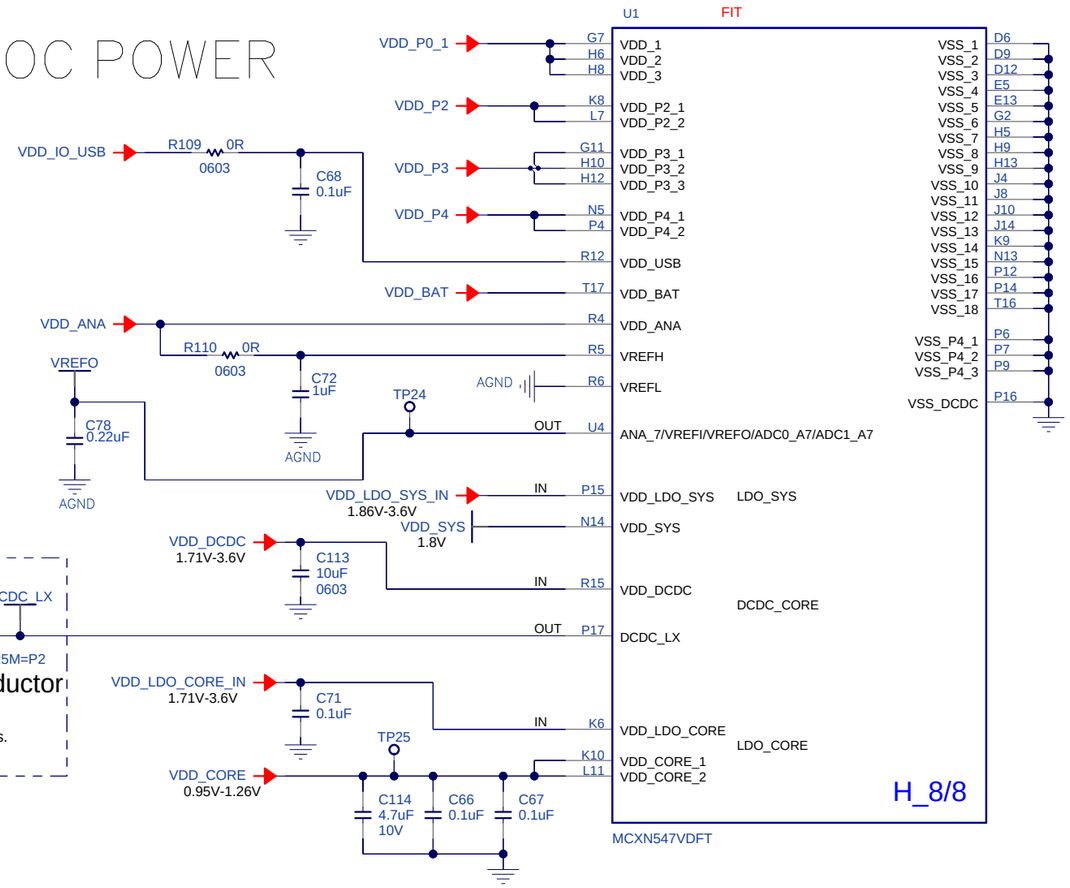


Layout notes:
======
Place these caps close to MCU pins.

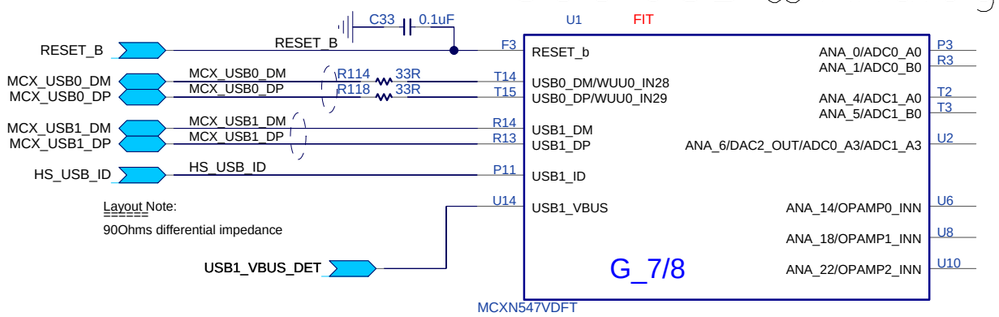




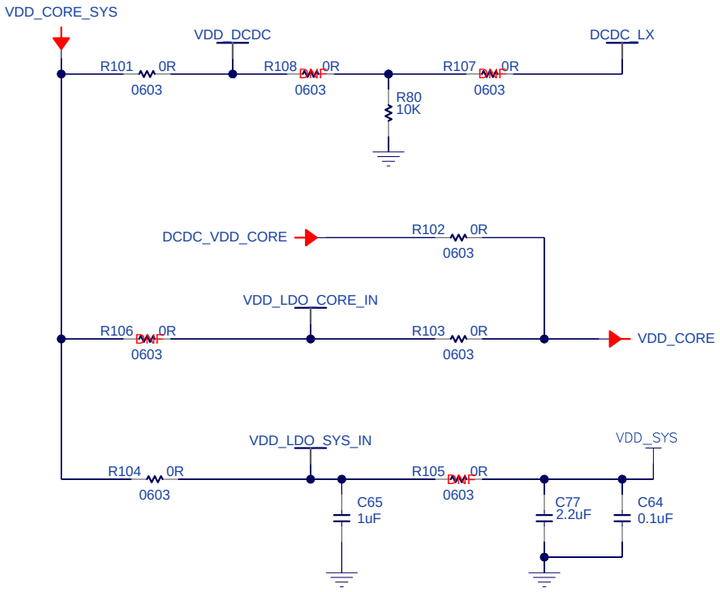
Layout notes:
======
1.Keep trace lengths short and components close to pins.
2.Place inductor and cap close to MCU



SOC USB & Analog Signals



Layout Note:
======
90Ohms differential impedance



DNP R105(default): Enable internal LDO_SYS.
Solder R105: Disable internal LDO_SYS.

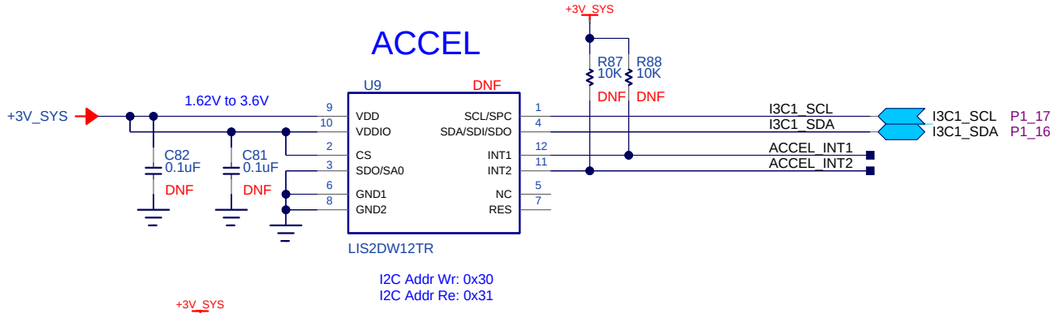
Mode	Solder Option
DCDC_CORE---Enable LDO_CORE---Disable (Default)	Populate R101, R102, R103; DNP R107, R108, R106.
DCDC_CORE---Disable LDO_CORE---Enable	Populate R107, R108, R106; DNP R101, R102, R103.

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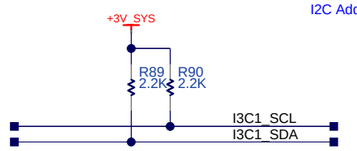
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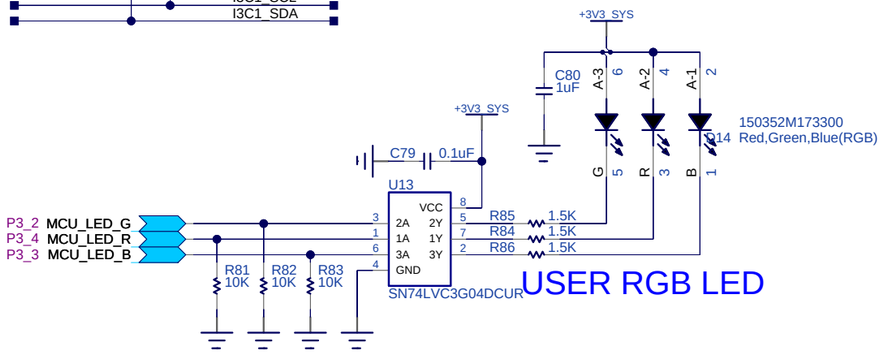
ACCEL



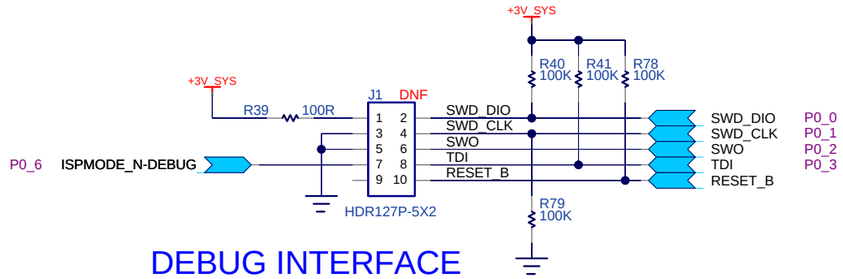
I2C Addr Wr: 0x30
I2C Addr Re: 0x31

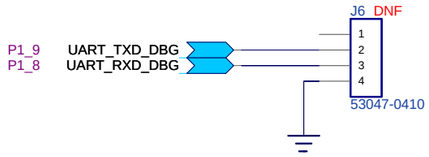


USER RGB LED

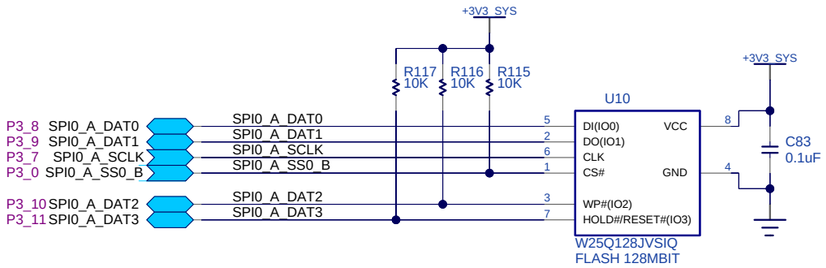


DEBUG INTERFACE





NOR FLASH



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